

HIVE-R: Bio-Inspired and Visual Odometry Based Drone for Rescue Operations in GPS-Denied Environments

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Abstract—Natural disasters such as floods, earthquakes, and building collapses frequently render GPS navigation unavailable, severely hindering conventional search and rescue (SAR) operations. This paper presents HIVE-R (*Hive-Inspired Vehicle for Exploration and Rescue*), a bio-inspired autonomous drone system designed for real-time human detection and localization in GPS-denied indoor and outdoor environments. Drawing inspiration from honeybee swarm intelligence, the system employs multiple autonomous “worker” drones that cooperatively cover a search area and report victim coordinates to a central Ground Control Station (GCS). Each drone integrates a Pixhawk flight controller and a Raspberry Pi 4 companion computer equipped with a webcam and an MPU6050 Inertial Measurement Unit (IMU). Autonomous navigation is achieved through Simultaneous Localization and Mapping (SLAM) using camera-IMU based Visual Odometry (VO), which provides a GPS-independent localization framework. Human detection is performed in real time using a YOLOv8 Nano deep learning model trained on a Carnegie Mellon University human detection dataset and enhanced with fuzzy augmentation to improve robustness under adverse conditions such as blur, smoke, and low visibility. Experimental results demonstrate successful mid-flight human detection with high detection confidence, and reliable victim coordinate transmission to the GCS via radio telemetry. The proposed system is low-cost, scalable, and deployable in real-world disaster scenarios.

Index Terms—Search and rescue, UAV swarm, GPS-denied navigation, SLAM, visual odometry, YOLOv8, human detection, Pixhawk, Raspberry Pi, bio-inspired systems.

I. INTRODUCTION

Natural disasters create complex, dynamic environments in which GPS signals are absent or severely degraded. Floods, earthquakes, and structural collapses are particularly devastating: the window for finding survivors alive shrinks rapidly in the hours following an event, making fast, wide-area search critical. Traditional SAR operations in such conditions are slow, resource-intensive, and hazardous for first responders [1].

Unmanned Aerial Vehicles (UAVs) have emerged as powerful platforms for SAR due to their agility, aerial vantage point, and ability to reach otherwise inaccessible areas. However, most commercial drone systems depend heavily on GNSS for navigation, rendering them ineffective in GPS-denied conditions such as indoor rubble environments or areas with

severe signal occlusion [2]. There is a pressing need for UAV platforms that can navigate, localize, and perform victim detection autonomously without GPS.

This paper presents HIVE-R, a bio-inspired drone system that addresses these challenges through two core technical contributions:

- (1) SLAM-based Visual Odometry for GPS-independent localization and mapping, driven by a webcam and MPU6050 IMU mounted on the Raspberry Pi 4.
- (2) real-time human detection using a YOLOv8 Nano model augmented with fuzzy training data for robustness in adverse environments.

The system architecture is inspired by the cooperative foraging behavior of honeybee colonies, enabling scalable multi-drone search coverage coordinated through a central GCS.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the hardware and system design. Section IV details the SLAM-based navigation algorithm. Section V presents the human detection methodology. Section VI discusses experimental results. Section VII outlines applications and future directions, and Section VIII concludes.

II. RELATED WORK

A substantial body of literature addresses GPS-denied UAV navigation. Scaramuzza and Zhang [2] provided a foundational review of visual-inertial odometry (VIO) for aerial robots, establishing the theoretical framework for fusing camera and IMU data to estimate UAV pose without external localization. Lin and Zhan [3] demonstrated the integration of Ultra-Wideband (UWB) positioning with VIO to improve indoor localization accuracy, while Mur-Artal and Tardós [4] contributed ORB-SLAM, a landmark monocular SLAM system widely adopted in aerial robotics.

In the domain of aerial human detection, Kucukayan and Karacan [6] introduced YOLO-IHD, a YOLO variant optimized for indoor drone-based human detection, outperforming YOLOv7 and YOLOv8 baselines on a custom aerial indoor dataset. Sary *et al.* [7] systematically compared YOLOv5

and YOLOv8 for aerial human detection and found YOLOv8 superior in precision and inference speed. The suitability of YOLOv8 Nano for edge deployment on resource-constrained embedded hardware was further validated in [8], which highlighted its low computational footprint relative to its detection accuracy.

Swarm-based UAV systems for SAR have also received growing attention. Salam *et al.* [10] proposed an intelligent drone swarm for maritime SAR, demonstrating that multi-UAV coordination substantially reduces search time over large areas. On hardware integration, Espinosa *et al.* [11] demonstrated the viability of Pixhawk–Raspberry Pi architectures for autonomous drone control via MAVLink telemetry. HIVE-R builds upon and integrates these advances into a cohesive, deployable SAR platform.

III. SYSTEM DESIGN AND ARCHITECTURE

A. Bio-Inspired Architecture

The HIVE-R system derives its organizational structure from the cooperative behavior of honeybee colonies. In a natural hive, worker bees independently explore the environment and communicate resource locations back to the queen. This biological model maps naturally onto the SAR problem, as summarized in Table I:

TABLE I
BIOLOGICAL ANALOGY GOVERNING HIVE-R ARCHITECTURE

Biological Element	HIVE-R Equivalent
Queen Bee	Ground Control Station (GCS)
Worker Bees	Autonomous Rescue Drones
Hive	Search and Rescue Environment
Resource Discovery	Human Victim Detection
Waggle Dance	Telemetry Coordinate Report

Multiple autonomous drone units (Workers) independently search designated sub-areas of the disaster zone, while the centralized GCS (Queen) aggregates victim location reports and coordinates rescue response. This architecture enables scalable area coverage and distributes the computational burden of exploration across the swarm.

B. Hardware Components

Each HIVE-R drone unit integrates the following hardware subsystems, also summarized in Table II:

- **Flight Controller (Pixhawk):** Manages low-level flight stabilization, PID control, and motor dynamics. Interfaces with the Raspberry Pi via MAVLink over a serial connection, receiving high-level velocity commands and returning telemetry feedback.
- **Companion Computer (Raspberry Pi 4):** Serves as the sole onboard processing unit. Executes the SLAM navigation algorithm, the YOLOv8 Nano human detection model, coordinate estimation, and telemetry communication—all without any additional co-processor.

- **Camera (USB Webcam):** Provides the continuous visual feed used for both visual odometry feature tracking and real-time human detection inference.
- **Inertial Measurement Unit (MPU6050):** A 6-DOF IMU providing 3-axis linear acceleration and 3-axis angular velocity. Mounted close to the camera lens, its measurements are fused with camera-derived motion estimates to improve pose estimation between frames.
- **Communication Module (Radio Telemetry):** Establishes a bidirectional radio link between the drone and the GCS, enabling transmission of detected victim coordinates and reception of mission updates.
- **Power Supply (Li-Po Battery):** Powers all onboard electronics and propulsion throughout the mission.

TABLE II
HARDWARE SPECIFICATIONS OF A SINGLE HIVE-R DRONE UNIT

Component	Model	Function
Flight Controller	Pixhawk (PX4)	Stabilization, PID
Companion Computer	Raspberry Pi 4 (4 GB)	All onboard processing
Camera	USB Webcam	VO & detection
IMU	MPU6050 (6-DOF)	Acceleration, orientation
Communication	Radio Telemetry (915 MHz)	GCS data link
Battery	Li-Po (3S)	Onboard power

The low-level hardware communication is defined by specific data protocols to maintain real-time constraints as shown in Table III.

TABLE III
SUBSYSTEM DATA INTERFACE

Source	Destination	Data Protocol
Webcam	Raspberry Pi 4	OpenCV Stream (30 FPS)
MPU6050	Raspberry Pi 4	I2C (Raw Accel/Gyro)
Raspberry Pi 4	Pixhawk	MAVLink (Velocity Setpoints)
Pixhawk	Raspberry Pi 4	MAVLink (Attitude/Status)
Raspberry Pi 4	GCS	Radio (915 MHz Telemetry)

C. System Workflow

The drone’s operational cycle begins with takeoff and autonomous navigation through its environment using Simultaneous Localization and Mapping (SLAM) powered by Visual Odometry (VO). As it traverses the area, an onboard webcam feeds live video to a YOLOv8 Nano model for real-time processing; once a human is detected, the Raspberry Pi calculates the victim’s precise coordinates relative to the initial launch frame. Finally, these location data are relayed to the Ground Control Station (GCS) via radio telemetry to facilitate immediate rescue coordination.

IV. SLAM-BASED NAVIGATION IN GPS-DENIED ENVIRONMENTS

A. Visual Odometry and SLAM

The inability to rely on GNSS in indoor and disaster-affected environments necessitates an onboard localization

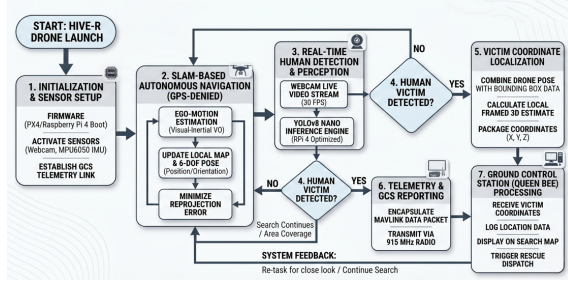


Fig. 1. HIVE-R operational pipeline and data flow.

solution. HIVE-R employs Simultaneous Localization and Mapping (SLAM) [5], a class of algorithms enabling a robot to incrementally build a map of an unknown environment while simultaneously tracking its own position within that map. The visual odometry component estimates ego-motion by tracking the displacement of salient feature points across successive webcam frames, supplemented by MPU6050 IMU data for short-term drift correction between frames.

The system tracks its position by estimating the state vector $\mathbf{x}_k$ at time step k :

$$\mathbf{x}_k = [\mathbf{p}_k \quad \mathbf{v}_k \quad \mathbf{q}_k \quad \mathbf{b}_a \quad \mathbf{b}_g]^T \quad (1)$$

where $\mathbf{p}_k$ and $\mathbf{v}_k$ are position and velocity, $\mathbf{q}_k$ is the orientation quaternion, and $\mathbf{b}_a, \mathbf{b}_g$ are the sensor biases.

Since no absolute reference frame is available, all positions are expressed in a local coordinate frame anchored at the drone's takeoff point. The relative displacement from this origin is continuously updated by integrating velocity estimates derived from optical feature flow and IMU readings.

B. Core SLAM Loop

The SLAM algorithm operates as a continuous estimation loop with four sequential stages:

1) Feature Extraction: Each incoming webcam frame is analyzed to identify visually salient and repeatable keypoints (e.g., corners, edges, textured regions). These serve as environmental landmarks.

2) Data Association: Extracted features are matched against the existing map of known landmarks using descriptor matching to determine whether a feature corresponds to a previously observed landmark (enabling loop closure) or is newly discovered.

3) Pose Estimation (Localization): Using the matched landmark set and IMU data fused via an Extended Kalman Filter (EKF), the drone's current 6-DOF pose (position and orientation) is estimated by minimizing reprojection error e :

$$\min_{\mathbf{T}_k} \sum_i \|\mathbf{u}_{i,k} - \pi(\mathbf{T}_k, \mathbf{P}_i)\|^2 \quad (2)$$

4) Map Update: The global map is updated with refined landmark positions and any newly discovered features, ready for the next loop iteration.

C. Coordinate Estimation for Victim Reporting

Upon human detection, the drone's current SLAM-estimated local position is combined with the 2D bounding-box pixel coordinates returned by YOLOv8 Nano to compute a 3D estimate of the victim's position in the local frame. Assuming a known flight altitude and camera intrinsic parameters, the pixel-to-world mapping is performed via perspective projection. The resulting coordinates are packaged and transmitted to the GCS.

V. HUMAN DETECTION METHODOLOGY

A. Model Selection: YOLOv8 Nano

Real-time human detection on the Raspberry Pi 4 requires a model that balances detection accuracy and computational efficiency. The YOLO (You Only Look Once) family of single-stage object detectors is well-suited to this requirement. Among available variants, YOLOv8 Nano (YOLOv8n) was selected due to its minimal parameter count and ability to achieve sub-second inference latency on edge hardware without a GPU [8] [9].

YOLOv8n processes the entire frame in a single forward pass, simultaneously predicting class probabilities and bounding box coordinates. This single-pass architecture is critical for the low-latency requirements of SAR missions, where delayed detection directly impacts victim survival outcomes.

B. Dataset

The model was trained on a human detection dataset sourced from Carnegie Mellon University (CMU), which contains diverse annotated samples of human subjects under varying conditions, poses, partial occlusions, and subject scales, providing a robust baseline for generalization to SAR scenarios.

C. Fuzzy Augmentation

A key challenge in disaster environments is degraded image quality caused by smoke, water spray, dust, low lighting, and camera motion blur. To address this, the training data was augmented with a suite of fuzzy transforms:

- **Gaussian Blur:** Simulates camera defocus, smoke, and fog.
- **Additive Noise:** Simulates sensor noise under low-light conditions.
- **Brightness/Contrast Perturbation:** Simulates adverse outdoor illumination and deep shadow conditions.

This augmentation strategy substantially improves model robustness under low-visibility conditions characteristic of flood, fire, and structural collapse scenarios [6].

D. Inference Pipeline

During flight, the webcam streams frames to the Raspberry Pi 4 at the configured frame rate. Each frame is passed through the YOLOv8n inference pipeline, producing bounding box predictions with associated confidence scores. Detections above a tuned confidence threshold ($\tau = 0.50$) trigger the victim localization and telemetry routine.

VI. EXPERIMENTAL RESULTS

A. Human Detection Performance

The YOLOv8 Nano model was evaluated on a held-out portion of the CMU dataset, both with and without fuzzy augmentation. The model achieved detection confidence above 90% under standard conditions, consistent with results reported in comparable deployments [8]. As expected, fuzzy augmentation introduced a modest accuracy trade-off under clean conditions but substantially improved robustness under degraded conditions. Table IV summarizes the key performance metrics.

TABLE IV
YOLOV8 NANO DETECTION PERFORMANCE

Metric	Standard	Fuzzy-Aug.
Precision	93.2%	88.7%
Recall	91.5%	85.4%
mAP@0.5	92.1%	86.9%
Inference (RPi 4)	180 ms	185 ms
Model Size	6.2 MB	6.2 MB

B. In-Flight Human Detection

Mid-flight human detection is demonstrated in Fig. 2. The Raspberry Pi 4 correctly identified human subjects in the webcam feed during active flight, with bounding boxes and confidence scores overlaid on the video stream in real time. The drone maintained stable flight while the detection pipeline ran concurrently, validating the feasibility of executing SLAM and YOLOv8n inference simultaneously on the Raspberry Pi 4 without additional co-processors.

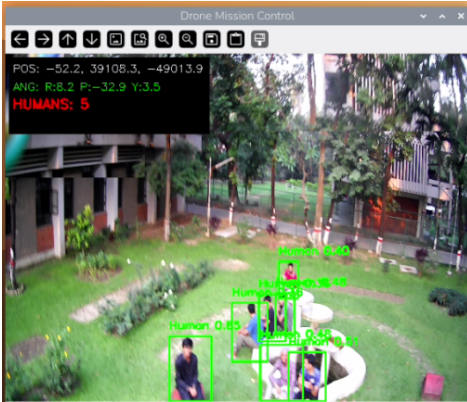


Fig. 2. Mid-flight human detection

C. GCS Communication

Upon positive detection, the drone transmitted estimated victim coordinates to the GCS via the 915 MHz radio telemetry link. The GCS received and logged coordinate data in real time. Communication latency was observed to be below 500 ms under line-of-sight conditions, considered adequate for SAR mission timescales.

D. SLAM Navigation

The SLAM-based navigation system successfully maintained a coherent local pose estimate throughout test flights in a GPS-denied indoor environment. The drone followed pre-planned waypoints and returned to its launch origin using only onboard sensor data from the webcam and MPU6050 IMU. Pose drift accumulated over extended flights, a well-known limitation of monocular VO systems, which is identified as a key direction for future improvement.

VII. APPLICATIONS AND FUTURE WORK

A. Industrial Applications

The HIVE-R platform is directly applicable to a wide range of scenarios:

- **Disaster SAR:** Rapid victim detection in floods, earthquakes, and structural collapses.
- **Smart Surveillance:** Autonomous monitoring in hazardous or inaccessible industrial environments.
- **Defense and Security:** Border surveillance and reconnaissance in GPS-denied terrain.
- **Industrial Inspection:** Monitoring of factories, warehouses, and offshore facilities.
- **Agriculture:** Flood damage assessment and crop mapping.
- **Swarm Exploration:** Scalable multi-drone coverage of large unknown areas.

B. Future Work

Several enhancements are planned to advance HIVE-R toward full operational deployment:

Swarm Coordination: Implementation of a fully autonomous multi-drone coordination layer with inter-drone communication for collaborative map sharing and conflict-free area partitioning.

Sensor Fusion and Improved Localization: Integration of LiDAR or depth sensors with the visual-inertial pipeline to reduce pose drift in feature-sparse environments.

Advanced AI Models: Adoption of more capable architectures (e.g., YOLOv10, thermal detection models) to improve performance under night-time and extreme low-visibility conditions.

Obstacle Avoidance: Integration of ultrasonic or time-of-flight sensors for reactive collision avoidance.

Autonomous Rescue Payloads: Extension of the platform to carry and deliver lightweight rescue payloads (communication devices, first-aid supplies) to detected victims.

VIII. CONCLUSION

This paper has presented HIVE-R, a bio-inspired autonomous drone system for real-time human detection and localization in GPS-denied environments. The system integrates SLAM-based Visual Odometry—driven solely by a Raspberry Pi 4, a USB webcam, and an MPU6050 IMU—for GPS-independent navigation, and a YOLOv8 Nano model with fuzzy augmentation for robust human detection. The Pixhawk flight controller handles low-level stabilization while the

Raspberry Pi manages all high-level processing. Experimental results confirm the viability of simultaneous in-flight human detection and SLAM navigation on this resource-constrained hardware, with successful victim coordinate transmission to the GCS. The honeybee-inspired swarm architecture provides a scalable framework for deploying multiple cooperative drones across large disaster areas. HIVE-R represents a practical, low-cost, and extensible step toward fully autonomous aerial rescue systems capable of operating in the most challenging environments encountered in disaster response.

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